

■ 4.1.8 Programming by Constraint Propagation

One important feature of transformation rules in *Mathematica* is that they are *directed*. You give a pattern, and specify how it should be transformed.

For some kinds of problems, a better approach is to specify a set of *constraints* that quantities should satisfy. Making an *assignment* like $x = y + 5p$ specifies that x should be *replaced* by $y + 5p$ whenever it appears. On the other hand, the *equation* $x == y + 5p$ merely represents the *constraint* that x should be equal to $y + 5p$.

One way to create a “program” is simply to build up a collection of constraints between different quantities. Then you can find values of quantities consistent with these constraints by explicitly using `Solve`.

```
va == vo r1 / (r1 + r2)
```

```
va == vi
```

```
g == vo / vi
```

Some equations giving constraints between various quantities.

Here are the constraint equations.

```
In[1]:= eqns = {
      va == vo r1 / (r1 + r2) ,
      va == vi ,
      g == vo / vi
    }
```

```
Out[1]= {va ==  $\frac{r1\ vo}{r1 + r2}$ , va == vi, g ==  $\frac{vo}{vi}$ }
```

This solves the equations for g in terms of $r1$ and $r2$.

```
In[2]:= Solve[eqns, g, {r1, r2}]
```

```
Out[2]= {{g ->  $\frac{vo}{va}$ }}
```

This solves for $r1$ in terms of g .

```
In[3]:= Solve[eqns, r1, g]
```

```
Out[3]= {{r1 ->  $-\left(\frac{r2\ va}{va - vo}\right)$ }}
```

If you set up a calculation using assignments, you are essentially giving a specific set of “inputs”, from which you compute a specific set of “outputs”. When you use equations, instead of assignments, you are giving relations with no definite “direction”. It is only when you use `Solve` to find the value of a particular variable that you are imposing a definite direction.

Setting up calculations using equations, rather than assignments, is particularly

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convenient if you need to “work backwards” from your output, to find appropriate “inputs”. You can use `Solve` equally well to find the “outputs” in terms of the “inputs”, or the “inputs” in terms of the “outputs”.

Building up a large collection of equations in *Mathematica* is something like setting up a spreadsheet. Each one of your variables corresponds to the value in a cell of the spreadsheet. Your equations specify the relations between values in different cells. You can use `Solve` to find the value of *any* cell in terms of others.

Whether an arrangement like this really works depends on how complicated the relations between different variables are. Essentially all traditional spreadsheet programs require all the relations to be *linear*. In this case, it is always possible to find a formula for the value of a particular variable, so long as the value is uniquely determined.

Mathematica allows you to give more complicated relations between variables. You can introduce non-linear dependencies by giving polynomial equations for one variable in terms of others. With equations like these, you cannot always get explicit formulae for one variable in terms of others.